

SECTION A

Answer all questions in the spaces provided.

1. When aqueous iodide ions are added to an aqueous solution of lead(II) nitrate, a precipitate is formed.

(a) Give the colour of the precipitate. [1]

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(b) Write an **ionic** equation for the reaction that occurs, including state symbols. [1]

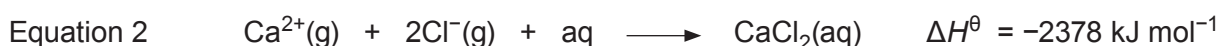
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2. Give an example of a transition metal used as a catalyst. You should name the metal and identify the reaction it catalyses. [1]

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3. Calcium chloride, CaCl_2 , is soluble in water. Use the values below to explain why this is the case. [2]



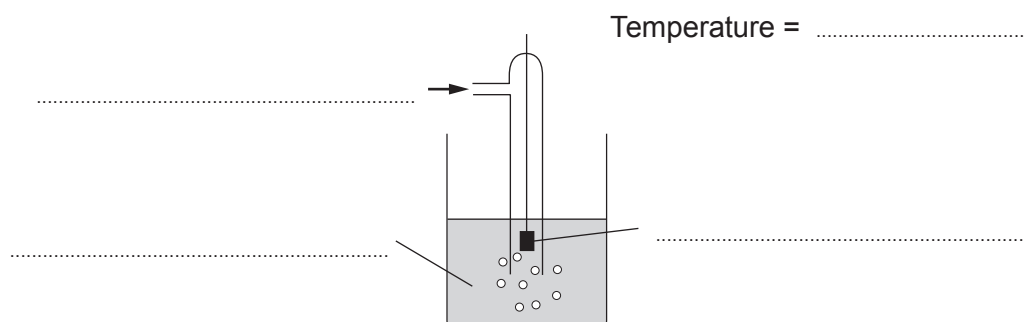
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4. Label the diagram of the standard hydrogen electrode below, including all relevant conditions. [2]



5. Reaction rates can be measured using an iodine clock reaction. The data below shows the results obtained by a student in an iodine clock experiment. The concentration of iodide ions was the only difference between the experiments.

Concentration of iodide, $[I^-] / \text{mol dm}^{-3}$	Time taken for colour change / s				Mean time taken / s
	1 st run	2 nd run	3 rd run	4 th run	
0.200	35	46	36	34	
0.400	18	17	18	17	

- (a) Complete the table with appropriate values of mean time taken. [1]
- (b) Calculate the rate of reaction for these two concentrations. [1]

Rate for $0.200 \text{ mol dm}^{-3} [I^-] = \dots\dots\dots \text{s}^{-1}$

Rate for $0.400 \text{ mol dm}^{-3} [I^-] = \dots\dots\dots \text{s}^{-1}$



6. Place the following substances in order of increasing entropy under standard conditions. [1]

nitrogen gas

copper metal

water

air

Lowest *Highest*

Examiner
only

10

